# COMPARATIVE SURVEY OF NEURAL NETWORK ARCHITECTURES

**Analyzing different neural network models and designs**

# TITLE AND OVERVIEW

### Comparing Neural Architectures

In the research I dived into surveys for CNNs, RNNs, ViTs, and hybrid models, highlighting their design and applications usage.

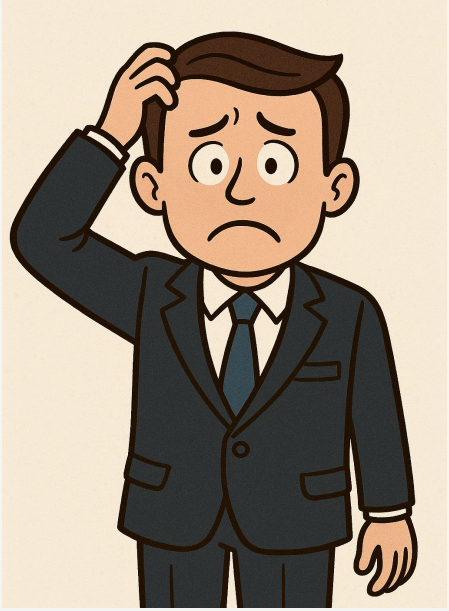
### Performance Factors

Emphasizes the importance of accuracy, efficiency, and flexibility in choosing a neural network architecture.

### Finally, Hybrid Models usage

Hybrid models combine strengths of CNNs and Transformers to handle complex tasks with local and global context.

# WHY THIS SURVEY MATTERS



## CNNs for Local Features

CNNs are effective to capture local spatial features, great in image classification especially with limited data availability.

## RNNs for Sequential Data

RNNs are designed for sequential data processing but face challenges scaling to large datasets or long sequences.

## Vision Transformers

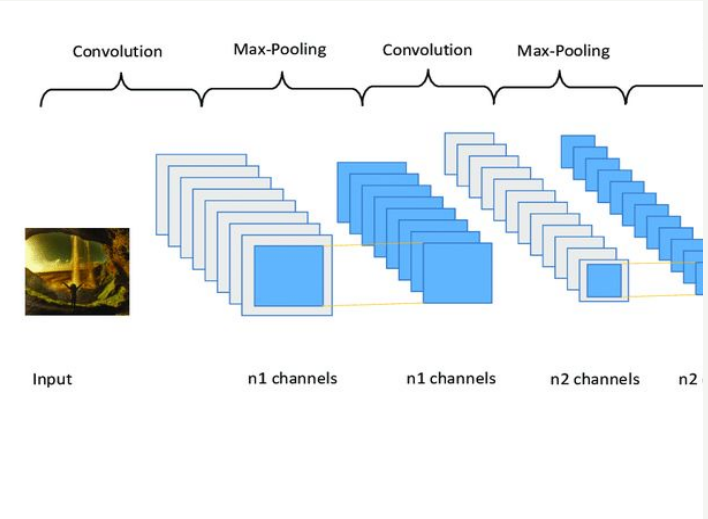
Vision Transformers use global attention mechanisms providing strong performance on large datasets with heavy computational demands.

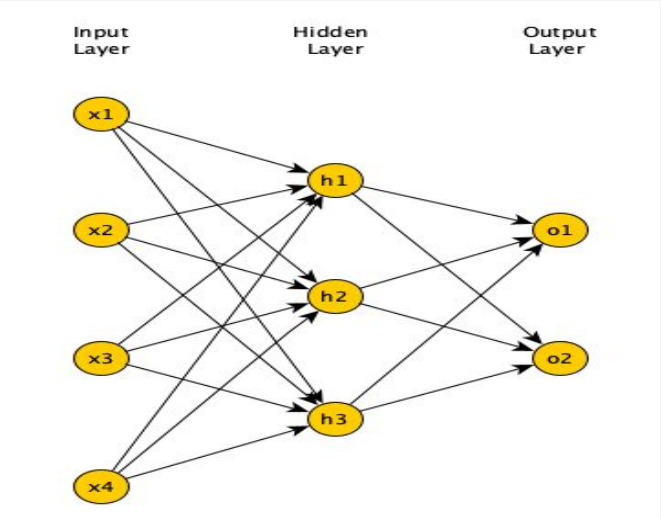
## Hybrid Neural Models

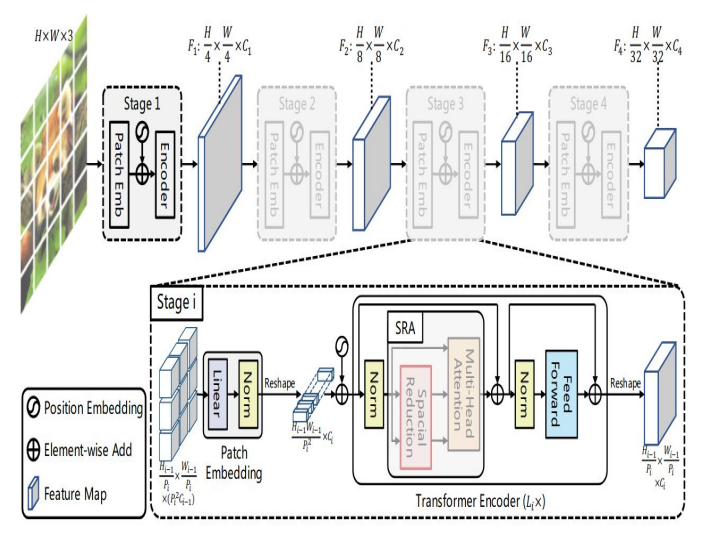
Hybrid models combine CNNs' local feature extraction with Transformer's global reasoning for balanced, versatile applications.

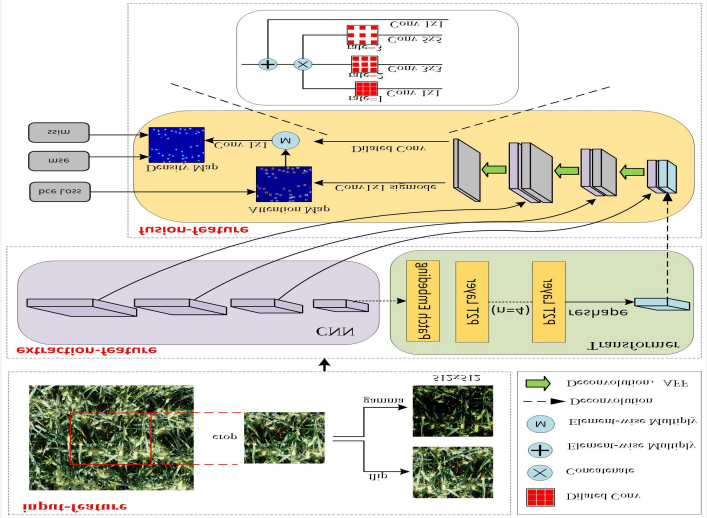
# ARCHITECTURAL LANDSCAPE AND FOUNDATIONS

# THE LANDSCAPE AT A GLANCE



## Convolutional Neural Networks (CNNs)

CNNs excel in computer vision with spatial locality and translation invariance biases for efficient image analysis.

## Recurrent Neural Networks (RNNs)

RNNs handle sequence data by maintaining temporal memory through recurrent connections for modeling time dependencies.

## Vision Transformers

Vision Transformers use global attention to capture complex and long-range relationships in visual data effectively.

## Hybrid Architectures

Hybrid models combine CNNs and Transformers to balance local feature extraction with global contextual understanding.

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# MATHEMATICAL FOUNDATIONS (INTUITIVE)

### Convolutional Neural Networks

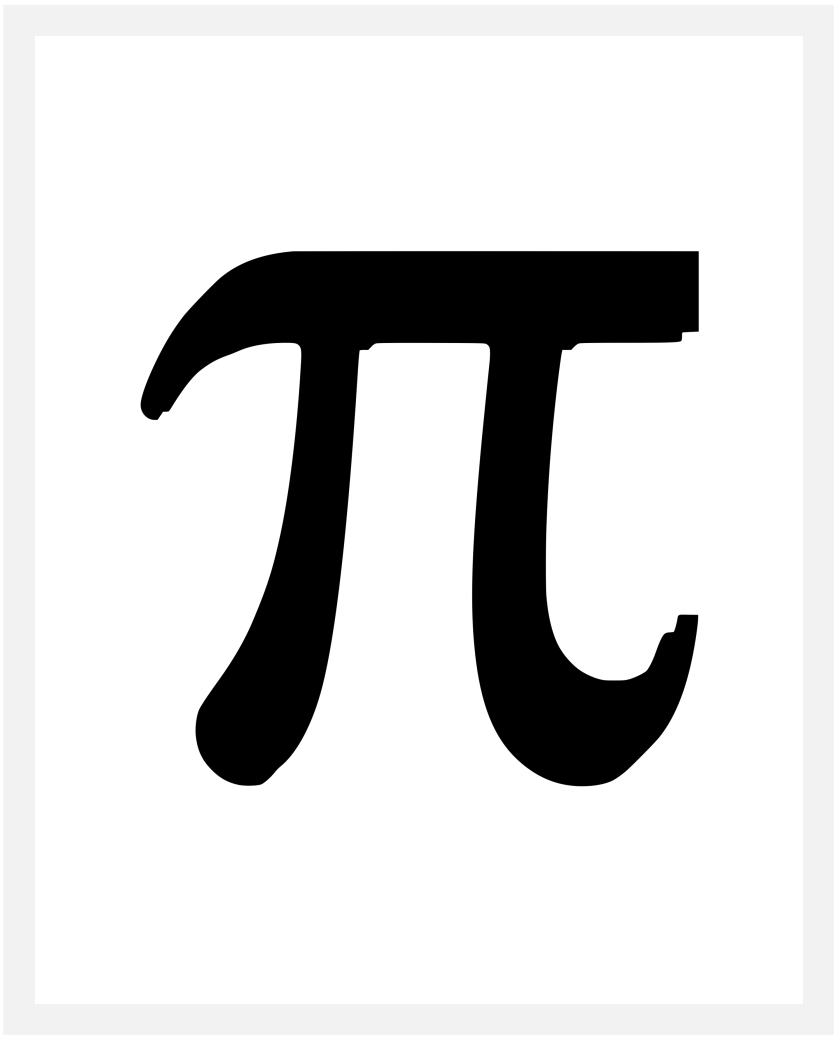
CNNs use convolutional filters to capture local spatial hierarchies in input data effectively.

### Recurrent Neural Networks

RNNs have a built-in memory that travels through the sequence, allowing them to understand how earlier inputs relate to later ones.

### Vision Transformers

Transformers use self-attention to model global relationships by computing pairwise relevance scores between tokens.



# ARCHITECTURES IN PRACTICE

# CNN IN PRACTICE

## Effectiveness in Image Tasks

CNNs capture local spatial patterns and maintain translation invariance, making them ideal for image-based tasks.

## Computational Efficiency

CNNs are computationally efficient and perform well on smaller datasets compared to attention-based models.

## Limitations in Global Context

CNNs struggle with modeling global context unless deeper or dilated convolutions are used.

## Common Applications

CNNs are widely used in image classification, medical imaging, and object detection tasks.

# RNN IN PRACTICE

## Sequential Data Processing

RNNs are designed to process sequential data like natural language and time-series effectively.

## LSTM and GRU Variants

LSTM (Long short-term memory) and GRU (Gated Recurrent Unit) models improve memory by using gating mechanisms to handle short-term dependencies.

## Training Challenges

RNNs suffer from vanishing and exploding gradients, making long sequence training difficult.

## Scalability Limitations

RNNs work step-by-step, so they can't process many parts of a sequence at the same time. This makes them slower and harder to scale compared to models that use attention, which can handle everything in parallel.

# VISION TRANSFORMERS IN PRACTICE

## Self-Attention Mechanisms

Vision Transformers use self-attention on image patches to model global context and learn versatile image representations.

## Scalability and Pretraining

ViTs perform well with large datasets and extensive pretraining, excelling in image recognition and multimodal tasks.

## Computational Challenges

Vision Transformers require high memory and processing power, making them computationally expensive and data-hungry.

## Optimization Techniques

Methods like pruning and distillation help make Vision Transformers smaller and faster, so they're easier to use in real-world applications.

# HYBRIDS: THE BEST OF BOTH WORLDS

## Local and Global Feature Integration

Hybrid models combine CNNs for detailed local features with Transformers for capturing global context effectively.

## Applications in Complex Domains

These architectures improve results in medical imaging and video analysis by addressing fine details and long-range dependencies.

## Challenges and Design Considerations

Hybrids face complexity, optimization issues, and require careful fusion and tokenization strategies during design.

## Balancing Accuracy and Flexibility

Managing computational constraints is key to leveraging hybrids for high accuracy and adaptable AI solutions.

# IMPLEMENTATION AND COMPARISON

# CODE REALITY (PYTORCH)

## CNN Architecture

CNNs use Conv2d, MaxPool, and Linear layers to perform image classification tasks efficiently.

## RNN Modules

RNNs use modules like RNN, LSTM, and GRU with linear layers to handle sequence data and generate outputs.

## Transformer Components

Transformers stack TransformerEncoderLayer and multi-head attention blocks to build deep, effective architectures.

## Code and Prototyping

PyTorch enables translating mathematical models into code for rapid prototyping and experimentation.

# COMPARISON TABLE

| ARCHITECTURE | STRENGTHS | WEAKNESSES | APPLICATIONS |
|---|---|---|---|
| CNN | Local features, data efficient | Weak global context | Image classification, medical imaging |
| RNN | Temporal modeling | Gradient issues, scale limits | NLP, speech, time-series |
| ViT | Global attention, flexible | Data-intensive, high compute | Large-scale vision, multimodal |
| Hybrid | Local + global | Complex to optimize | Medical imaging, video/action |

# APPLICATIONS, CHALLENGES, AND CONCLUSIONS

# EVIDENCE FROM DOMAINS

## Medical Imaging Performance

CNNs perform well in medical imaging tasks with limited labeled data, outperforming Vision Transformers in many cases.

## Hybrid Models Advantages

Hybrid architectures combine strengths for tasks needing global context across multiple slices or views in medical imaging.

## Action Recognition Roles

CNNs manage spatial frame processing while Transformers take temporal and global relationships in action recognition.

## Impacting Factors

Model performance depends on data availability, task complexity, and computational resources across domains.

# CHALLENGES AND GAPS

## Data Efficiency Challenges

Data efficiency is vital, prompting advances in few-shot and self-supervised learning research.

## Computational Constraints

Computational and memory limits require efficient attention mechanisms, pruning, and distillation techniques.

## Optimization and Tokenization

Stable optimization for hybrid models and effective tokenization strategies need careful design.

## Interpretability Concerns

Interpretability techniques like saliency maps and attention visualization provide partial insights.

# PRACTICAL SELECTION GUIDE

## CNNs for Small Datasets

Convolutional Neural Networks are good with small datasets when combined with transfer learning and regularization techniques.

## Vision Transformers Advantage

Vision Transformers perform well in capturing long-range context but require extensive pretraining resources.

## Hybrid Models Use Case

Hybrid architectures balance local and global features, needing careful tuning for optimal performance.

## Efficient Deployment

Model efficiency through quantization and distillation is key for deployment under compute constraints.

# CONCLUSIONS

## Architectural Diversity

Different architectures have unique inductive biases, offering various strengths and limitations based on context.

## Hybrid Models

Hybrid architectures combine local and global capabilities but increase system complexity and design challenges.

## Future Research Directions

Improving efficiency, interpretability, and robustness is essential for advancing architectural designs.

## Informed Model Selection

Selecting architectures wisely leads to better performance and optimal resource use across applications.